Muhammad Mursaleen

AI & ML Engineer

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Professional Summary

AI & ML Engineer with almost 3 years of experience in building production-grade LLM applications, RAG systems, and agent workflows. Delivered FastAPI-based AI platforms serving healthcare, finance, procurement and Islamic knowledge domains using LangGraph, Qdrant, AWS, and Docker. Experienced in fine-tuning LLM's, retrieval optimization , and scalable Gen-AI and backend development.

Experience

Software Engineer (AI/ML) **July 2026 – Present**
CodingCops Lahore, Pakistan

- Designed and built multi-agent AI systems and text-to-SQL pipelines using LangGraph and self-hosted LLMs, optimizing prompts, context management, and token usage to improve reasoning efficiency and handle complex natural-language queries across large datasets.

AI/ML Engineer **March 2025 – July 2026**
DiveDeepAI Islamabad, Pakistan

- Built and deployed Generative AI systems using RAG, LangGraph, LLM fine-tuning, FastAPI, AWS EC2, and DigitalOcean, while leading a 3-person engineering team and driving system architecture.

Jr AI Engineer **August 2024 – February 2025**
Softoo Islamabad, Pakistan

- Built production-oriented Generative AI POCs and RAG applications using LangChain, LangGraph, LLMs, and vector databases, working closely with solution architects to validate and iterate on AI products.

Projects

Bridge Vet: Veterinary Clinic Platform

- Managed a development team to deliver an end-to-end clinic management system; built a scalable FastAPI, Redis, and Celery backend for asynchronous AI SOAP note generation, medication extraction, and appointment follow-ups.
- Developed a custom RAG-based clinical assistant for dosage calculations, evaluated retrieval accuracy via Langfuse, and implemented OCR pipelines to automate inventory tracking. .

Tech Stack: Python, FastAPI, Redis, Celery, Gemini, Vector Databases, Langfuse, OCR, AsyncIO, Docker, AWS EC2

eQRA-AI: Islamic AI Assistant Mobile App

- Built the backend for an Islamic AI Assistant, deploying dual RAG chatbots dedicated to precise Fatwa and Duas retrieval with Mem0-powered long-term conversational memory; engineered an event-driven notification system using AWS EventBridge, SNS, and Lambda to trigger real-time push notifications via FCM, and scaled deployments on DigitalOcean App Platform with Redis/Celery over droplets and self hosted Grafana Loki for log aggregation.

Tech Stack: Python, FastAPI, Mem0, AWS EventBridge, AWS SNS, AWS Lambda, FCM, Redis, Celery, Grafana Loki, DigitalOcean

Yani: AI Assistant for Chartered Accountants

- Built a RAG-based chatbot to assist Chartered Accountants, alongside a distinct document compliance workflow that evaluates text chunks directly against regulatory frameworks to map clauses and flag non-compliance.
- Implemented an AI-powered database chat utilizing Vanna.ai trained on complex schemas, achieving 92% accuracy in natural language to SQL conversion.
- Developed a multi-agent system using Autogen to perform automated Exploratory Data Analysis (EDA) and dynamically generate visual insights from the extracted SQL data.

Tech Stack: Python, Vanna.ai, Autogen, RAG, LLMs, Pandas, Matplotlib, Streamlit

Fine-Tuning Whisper for Quranic Arabic Transcription

- Optimized a custom fine-tuning pipeline for the Whisper model utilizing 10,000+ Quranic Tarteel audio samples to improve specialized Arabic speech-to-text transcription and phonetic accuracy.
- Achieved a 90% improvement in complex Harakat (diacritic) recognition and reduced overall Word Error Rate (WER) by 30–40% compared to foundational baseline models.

Tech Stack: Python, PyTorch, Hugging Face, Whisper, Audio Processing

Education

The Islamia University of Bahawalpur **Punjab, Pakistan**
Bachelor of Science in Software Engineering; CGPA: 3.70/4.00 Oct. 2020 – June 2024

Skills

Languages: Python, SQL
AI/ML & Frameworks: PyTorch, Hugging Face, OpenAI , LangChain, LangGraph, Autogen, RAG, Mem0
Evaluation & Observability: Langfuse, Grafana Loki, LLM Benchmarking, Retrieval Evaluation, Hallucination Reduction
Databases: PostgreSQL, Vector DBs (Qdrant, FAISS, PgVector), Redis, MongoDB
Backend & Infrastructure: FastAPI, Celery, AWS (EC2, S3, EventBridge, SNS, Lambda), DigitalOcean, Docker, Linux, CI/CD
Tools: Git, GitHub, GitLab, Web Scraping, Postman, Claude Code , Cursor , Antigravity.